

JEE (Main)-2026 Session-1
Question Paper with Solutions
(Mathematics, Physics, And Chemistry)
24 January 2026 Shift – 2

Time: 3 hrs.

M.M: 300

IMPORTANT INSTRUCTIONS:

- (1) The test is of 3 hours duration.
- (2) This test paper consists of 75 questions. Each subject (PCM) has 25 questions. The maximum marks are 300.
- (3) This question paper contains Three Parts. Part-A is Physics, Part-B is Chemistry and Part-C is Mathematics. Each part has only two sections: Section-A and Section-B.
- (4) Section - A: Attempt all questions.
- (5) Section - B: Attempt all questions.
- (6) Section - A (01 - 20) contains 20 multiple choice questions which have only one correct answer. Each question carries +4 marks for correct answer and -1 mark for wrong answer.
- (7) Section - B (21 - 25) contains 5 Numerical value-based questions. The answer to each question should be rounded off to the nearest integer. Each question carries +4 marks for correct answer and -1 mark for wrong answer.

MATHEMATICS

SECTION-A

1. Let $f(x) = \int \frac{7x^{10} + 9x^8}{(1+x^2+2x^9)^2} dx, \quad x > 0,$

$\lim_{x \rightarrow 0} f(x) = 0$ and $f(1) = \frac{1}{4}$. If

$A = \begin{bmatrix} 0 & 0 & 1 \\ \frac{1}{4} & f'(1) & 1 \\ \alpha^2 & 4 & 1 \end{bmatrix}$ and $B = \text{adj}(\text{adj } A)$ be such

that $|B| = 81$, then α^2 is equal to

- (1) 2 (2) 3
(3) 1 (4) 4

Ans. (4)

Sol. $f(x) = \int \left(\frac{7}{x^8} + \frac{9}{x^{10}} \right) dx$
 $\left(\frac{1}{x^9} + \frac{1}{x^7} + 2 \right)^2$

Put $t = \frac{1}{x^9} + \frac{1}{x^7} + 2 \Rightarrow \frac{dt}{dx} = \frac{-9}{x^{10}} - \frac{7}{x^8}$

$f(x) = \int \frac{-dt}{t^2} = \frac{1}{t} + C$

$f(x) = \frac{1}{\frac{1}{x^9} + \frac{1}{x^7} + 2} + C$

$= \frac{x^9}{1+x^2+2x^9} + C$

Given $f(1) = \frac{1}{4} = \frac{1}{4} + C \Rightarrow C = 0$

$f(x) = \frac{x^9}{1+x^2+2x^9}$

$f'(x) = \frac{(1+x^2+2x^9) - 9x^8 - x^9(2x+18x^8)}{(1+x^2+2x^9)^2}$

$f'(x) = \frac{36-20}{16} = 1$

$A = \begin{pmatrix} 0 & 0 & 1 \\ 4 & 1 & 1 \\ \alpha^2 & \frac{1}{4} & 1 \end{pmatrix}$

$|A| = |1 - \alpha^2| = 3$

$1 - \alpha^2 = 3, -3 \Rightarrow \alpha^2 = -2, 4$

Value of $\alpha^2 = 4$

$B = \text{adj}(\text{adj } A)$

$|B| = 81 = |A|^4 \Rightarrow |A| = 3$

2. Let the length of the latus rectum of an ellipse

$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, (a > b)$, be 30. If its eccentricity is the

maximum value of the function $f(t) = -\frac{3}{4} + 2t - t^2$,

then $(a^2 + b^2)$ is equal to -

- (1) 516 (2) 256
(3) 496 (4) 276

Ans. (3)

Sol. $f(t) = \frac{-3}{4} + 2t - t^2$

$f(t) \Big|_{\text{maximum}} = \frac{1}{4} = e \Rightarrow e^2 = \frac{1}{16} \Rightarrow \frac{a^2 - b^2}{a^2} = \frac{1}{16} \dots (1)$

$\therefore \frac{2b^2}{a} = 30 \Rightarrow b^2 = 15a \dots (2)$

By (1) & (2)

$16(a^2 - 15a) = a^2 \Rightarrow 15a^2 - 16 \times 15a = 0$

$a = 16$

$b^2 = 240$

$a^2 + b^2 = 256 + 240$

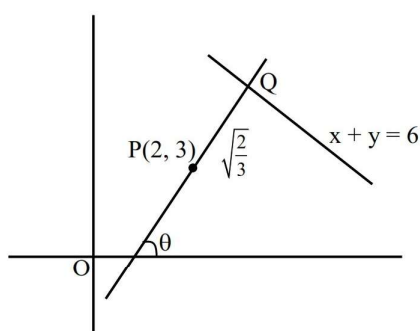
$= 496$

3. Let the angles made with the positive x-axis by two straight lines drawn from the point $P(2, 3)$ and meeting the line $x + y = 6$ at a distance $\sqrt{\frac{2}{3}}$ from the point P be θ_1 and θ_2 . Then the value of $(\theta_1 + \theta_2)$ is :

- (1) $\frac{\pi}{12}$ (2) $\frac{\pi}{6}$
 (3) $\frac{\pi}{2}$ (4) $\frac{\pi}{3}$

Ans. (3)

Sol.



Let Q is $\left(\sqrt{\frac{2}{3}} \cos \theta + 2, \sqrt{\frac{2}{3}} \sin \theta + 3\right)$

so, $x + y = 6$

$$\sqrt{\frac{2}{3}} (\cos \theta + \sin \theta) + 5 = 6$$

$$\sin \theta + \cos \theta = \sqrt{\frac{3}{2}}$$

$$1 + \sin 2\theta = \frac{3}{2}$$

$$\sin 2\theta = \frac{1}{2}$$

$$2\theta = \frac{\pi}{6}, \frac{5\pi}{6}$$

$$\theta = \frac{\pi}{12} \text{ \& } \frac{5\pi}{6}$$

So $\theta_1 + \theta_2 = \frac{\pi}{2}$

4. Let $\vec{a} = 2\hat{i} - \hat{j} - \hat{k}$, $\vec{b} = \hat{i} + 3\hat{j} - \hat{k}$ and $\vec{c} = 2\hat{i} + \hat{j} + 3\hat{k}$. Let $\vec{v}$ be the vector in the plane of the vectors $\vec{a}$ and $\vec{b}$, such that the length of its projection on the vector $\vec{c}$ is $\frac{1}{\sqrt{14}}$. Then $|\vec{v}|$ is equal to

- (1) $\frac{\sqrt{21}}{2}$ (2) 13
 (3) $\frac{\sqrt{35}}{2}$ (4) 7

Ans. (Bonus)

NTA (3)

Sol. $\vec{v} = x\vec{a} + y\vec{b} = x(2\hat{i} - \hat{j} - \hat{k}) + y(\hat{i} + 3\hat{j} - \hat{k})$

$$\vec{v} = (2x + y)\hat{i} + (3y - x)\hat{j} + (-x - y)\hat{k}$$

$$\frac{|\vec{v} \cdot \vec{c}|}{|\vec{c}|} = \frac{1}{\sqrt{14}}$$

$$\vec{v} \cdot \vec{c} = 2(2x + y) + 3y - x - 3x - 3y$$

$$= 2y$$

$$\left| \frac{2y}{\sqrt{14}} \right| = \frac{1}{\sqrt{14}} \Rightarrow |2y| = 1$$

$$|\vec{v}| = \sqrt{(2x + y)^2 + (3y - x)^2 + (-x - y)^2}$$

$$= \sqrt{6x^2 + 11y^2 + 4xy - 6xy + 2xy}$$

$$= \sqrt{6x^2 + \frac{11}{4}} = \frac{\sqrt{24x^2 + 11}}{2}$$

Now if we take $x^2 = 1$ then option $\frac{\sqrt{35}}{2}$ matches

most probably NTA thought could be this.

5. Let $\alpha_1, \alpha_2, \alpha_3, \alpha_4$ be an A.P. of four terms such that each term of the A. P. and its common difference l are integers. If $\alpha_1 + \alpha_2 + \alpha_3 + \alpha_4 = 48$ and $\alpha_1, \alpha_2, \alpha_3, \alpha_4 + l^4 = 361$ then the largest term of the A.P. is equal to

- (1) 27 (2) 24 (3) 21 (4) 23

Ans. (1)

Sol. a_1, a_2, a_3, a_4 as $a - 3d, a - d, a + d, a + 3d$

where $\boxed{d = \frac{\ell}{2}}$

$$\therefore a_1 + a_2 + a_3 + a_4 = 48 \Rightarrow 4a = 48 \Rightarrow a = 12$$

$$\& a_1 a_2 a_3 a_4 + \ell^4 = 361 \Rightarrow (a^2 - 9d^2)(a^2 - d^2) + 16d^4 = 361$$

$$\Rightarrow (144 - 9d^2)(144 - d^2) + 16d^4 = 361$$

$$\Rightarrow 25d^4 - 1440d^2 + (144)^2 = 361$$

$$(5d^2 - 144)^2 = 19^2$$

$$\therefore 5d^2 - 144 = 19 \text{ or } -19$$

$$d^2 = \frac{163}{5} \text{ or } d^2 = \frac{125}{5} = 25$$

$$d = \sqrt{\frac{163}{5}} \text{ or } d = 5$$

$$\therefore \ell = 2\sqrt{\frac{163}{5}} \text{ or } \ell = 10$$

(rejected)

$\therefore$ common difference is an integer

$$\therefore \text{largest term} = 12 + 15 = 27$$

6. Let the image of parabola $x^2 = 4y$, in the line $x - y = 1$ be $(y + \alpha)^2 = b(x - c)$, $a, b, c \in \mathbb{N}$. Then $a + b + c$ is equal to

(1) 12 (2) 4

(3) 6 (4) 8

Ans. (3)

Sol. Parametric point P on $x^2 = 4y$ is $P(2t, t^2)$

$\therefore$ mirror image of P in $x - y = 1$ is

$$Q \equiv \left(2t - \frac{2 \cdot 1 \cdot (2t - t^2 - 1)}{2}, t^2 + \frac{2 \cdot 2(1) \cdot (2t - t^2 - 1)}{2} \right)$$

$$Q \equiv (t^2 + 1, 2t - 1) \equiv (h, k)$$

$\therefore$ locus of Q is $x = \frac{(y+1)^2}{4} + 1$ which is the required parabola.

$$\therefore \boxed{(y+1)^2 = 4(x-1)}$$

$$\therefore a = 1, b = 4, c = 1$$

$$\therefore \boxed{a+b+c=6}$$

7. $\left(\frac{1}{3} + \frac{4}{7}\right) + \left(\frac{1}{3^2} + \frac{1}{3} \times \frac{4}{7} + \frac{4^2}{7^2}\right) +$
 $\left(\frac{1}{3^3} + \frac{1}{3^2} \times \frac{4}{7} + \frac{1}{3} \times \frac{4^2}{7^2} + \frac{4^3}{7^3}\right) + \dots$ upto infinite terms is equal to -

(1) $\frac{5}{2}$ (2) $\frac{7}{4}$ (3) $\frac{4}{3}$ (4) $\frac{6}{5}$

Ans. (1)

Sol. Let $a = \frac{4}{7}$, $b = \frac{1}{3}$

Multiply N^r and D^r by $(a - b) = \frac{4}{7} - \frac{1}{3} = \frac{5}{21}$

$$\frac{1}{a-b} [(a^2 - b^2) + (a^3 - b^3) + (a^4 - b^4) + \dots \infty]$$

$$\frac{1}{a-b} \left[\frac{a^2}{1-a} - \frac{b^2}{1-b} \right] = \frac{21}{5} \left[\frac{\frac{16}{49}}{1-\frac{4}{7}} - \frac{\frac{1}{9}}{1-\frac{1}{3}} \right]$$

$$= \frac{21}{5} \left[\frac{16}{21} - \frac{1}{6} \right] = \frac{21}{5} \left[\frac{96-21}{21 \cdot 6} \right]$$

$$= \frac{75}{5 \cdot 6} = \frac{15}{6} = \frac{5}{2}$$

8. Let $P = [p_{ij}]$ and $Q = [q_{ij}]$ be two square matrices of order 3 such that $q_{ij} = 2^{(i+j-1)} p_{ij}$ and $\det(Q) = 2^{10}$. Then the value of $\det(\text{adj}(\text{adj}(P)))$ is :

(1) 32 (2) 16
(3) 81 (4) 124

Ans. (2)

Sol. $\begin{vmatrix} 2p_{11} & 2^2 p_{12} & 2^3 p_{13} \\ 2^2 p_{21} & 2^3 p_{22} & 2^4 p_{23} \\ 2^3 p_{31} & 2^4 p_{32} & 2^5 p_{33} \end{vmatrix} = 2^{10}$

$$2^2 \cdot 2 \cdot 2^3 \begin{vmatrix} p_{11} & p_{12} & p_{13} \\ 2p_{21} & 2p_{22} & 2p_{23} \\ 2^2 p_{31} & 2^2 p_{32} & 2^2 p_{33} \end{vmatrix} = 2^{10}$$

$$2^9 \begin{vmatrix} p_{11} & p_{12} & p_{13} \\ p_{21} & p_{22} & p_{23} \\ p_{31} & p_{32} & p_{33} \end{vmatrix} = 2^{10} \Rightarrow |P| = 2$$

$$|\text{adj}(\text{adj}(P))| = |P|^{(n-1)^2} = |P|^4 = 2^4 = 16$$

9. The letters of the word "UDAYPUR" are written in all possible ways with or without meaning and these words are arranged as in a dictionary. The rank of the word "UDAYPUR" is :

- (1) 1580 (2) 1578
(3) 1579 (4) 1581

Ans. (1)

Sol. ADIPRUU

$$A \rightarrow \frac{6!}{2!} = 360$$

$$D \rightarrow \frac{6!}{2!} = 360$$

$$P \rightarrow \frac{6!}{2!} = 360$$

$$R \rightarrow \frac{6!}{2!} = 360$$

$$UA \rightarrow 5! = 120$$

$$UDAP \rightarrow 3! = 6$$

$$UDAR \rightarrow 3! = 6$$

$$UDAU \rightarrow 3! = 6$$

$$UDAYPRU \rightarrow 1$$

$$UDAYPUR \rightarrow 1$$

$$\text{Total} = 1580$$

10. The largest value of n , for which 40^n divides $60!$, is

- (1) 13 (2) 11
(3) 12 (4) 14

Ans. (4)

Sol. $40^n = 2^{3n} \times 5^n$

$$E_2(60!) = \left[\frac{60}{2} \right] + \left[\frac{60}{2^2} \right] + \left[\frac{60}{2^3} \right] + \left[\frac{60}{2^4} \right] + \left[\frac{60}{2^5} \right]$$

$$= 30 + 15 + 7 + 3 + 1 = 56$$

$$E_5(60!) = \left[\frac{60}{5} \right] + \left[\frac{60}{5^2} \right]$$

$$= 12 + 2 = 14$$

$$40^n = (2^3)^n \times 5^n = (2^3 \times 5)^n$$

$$60! = 2^{56} \times 5^{14} \dots = 2^{14} \cdot (2^3 \cdot 5)^{14}$$

$\therefore$ Maximum value of n is 14.

11. The sum of all values of α , for which the shortest distance between the lines

$$\frac{x+1}{\alpha} = \frac{y-2}{-1} = \frac{z-4}{-\alpha} \text{ and } \frac{x}{\alpha} = \frac{y-1}{2} = \frac{z-1}{2\alpha} \text{ is}$$

$$\sqrt{2}, \text{ is}$$

- (1) 8 (2) -6
(3) 6 (4) -8

Ans. (2)

$$\text{Sol. } \sqrt{2} = \frac{\begin{vmatrix} -1 & 1 & 3 \\ \alpha & -1 & -\alpha \\ \alpha & 2 & 2\alpha \end{vmatrix}}{\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \alpha & -1 & -\alpha \\ \alpha & 2 & 2\alpha \end{vmatrix}}$$

$$\sqrt{2} = \frac{-1(-2\alpha + 2\alpha) - 1(2\alpha^2 + \alpha^2) + 3(2\alpha + \alpha)}{\hat{i}(-2\alpha + 2\alpha) - \hat{j}(2\alpha^2 + \alpha^2) + \hat{k}(2\alpha + \alpha)}$$

$$\sqrt{2} = \frac{-3\alpha^2 + 9\alpha}{\sqrt{9\alpha^4 + 9\alpha^2}}$$

$$\sqrt{2} = \frac{-\alpha + 3}{\sqrt{\alpha^2 + 1}}$$

$$\Rightarrow 2\alpha^2 + 2 = \alpha^2 + 9 - 6\alpha$$

$$\alpha^2 + 6\alpha - 7 = 0$$

$$(\alpha + 7)(\alpha - 1) = 0$$

$$\alpha = -7, 1$$

$$\text{sum} = -7 + 1 = -6$$

option (2)

12. Let $f(\alpha)$ denote the area of the region in the first quadrant bounded by $x = 0$, $x = 1$, $y^2 = x$ and $y = |\alpha x - 5| - |1 - \alpha x| + \alpha x^2$. Then $(f(0) + f(1))$ is equal to

- (1) 9 (2) 14
(3) 7 (4) 12

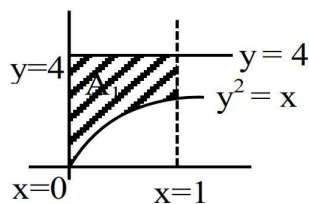
Ans. (3)

Sol. at $\alpha = 0 \Rightarrow f(0)$

$$x = 0, x = 1, y^2 = x$$

$$y = |0 \cdot x - 5| - |1 - 0 \cdot x| + 0 \cdot x^2$$

$$y = 4$$



$$A_1 = \int_0^1 (4 - \sqrt{x}) dx$$

$$= 4x - \frac{x^{\frac{3}{2}}}{\frac{3}{2}} \bigg|_0^1$$

$$= 4 - \frac{2}{3}(1) = \frac{10}{3}$$

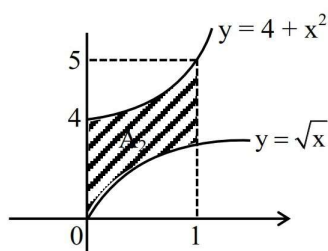
at $\alpha = 1 \Rightarrow f(1)$

$x = 0, x = 1, y^2 = x,$

$y = |x - 5| - |1 - x| + x^2$ in $x \in (0, 1)$

$y = 5 - x - (1 - x) + x^2$

$y = 4 + x^2$



$$A_2 = \int_0^1 ((4 + x^2) - (\sqrt{x})) dx$$

$$= 4x + \frac{x^3}{3} - \frac{x^{\frac{3}{2}}}{\frac{3}{2}} \bigg|_0^1$$

$$= 4 + \frac{1}{3} - \frac{2}{3} = \frac{11}{3}$$

$$|f(0) + f(1)| = |A_1 + A_2| = \left| \frac{10}{3} + \frac{11}{3} \right| = \left| \frac{21}{3} \right| = 7$$

option (3)

13. If the domain of the function $f(x) = \sin^{-1} \left(\frac{1}{x^2 - 2x - 2} \right)$, is $(-\infty, \alpha] \cup [\beta, \gamma] \cup [\delta, \infty)$, then

$\alpha + \beta + \gamma + \delta$ is equal to

(1) 2 (2) 4

(3) 3 (4) 5

Ans. (2)

Sol. $-1 \leq \frac{1}{x^2 - 2x - 2} \leq 1$

$$\frac{1 + x^2 - 2x - 2}{x^2 - 2x - 2} \geq 0 \Rightarrow \frac{(x-1)^2 - 2}{(x-1)^2 - 3} \geq 0$$

$$\Rightarrow \frac{(x-1-\sqrt{2})(x-1+\sqrt{2})}{(x-1-\sqrt{3})(x-1+\sqrt{3})} \geq 0$$

$$x \in (-\infty, 1-\sqrt{3}) \cup [1-\sqrt{2}, 1+\sqrt{2}] \cup (1+\sqrt{3}, \infty) \dots (1)$$

$$1 - \frac{1}{x^2 - 2x - 2} \geq 0 \Rightarrow \frac{x^2 - 2x - 3}{x^2 - 2x - 2} \geq 0$$

$$\Rightarrow \frac{(x+1)(x-3)}{(x-1+\sqrt{3})(x-1-\sqrt{3})} \geq 0$$

$$x \in (-\infty, -1] \cup (1-\sqrt{3}, \sqrt{3}+1) \cup [3, \infty) \dots (2)$$

(1) $\cap$ (2)

$$\Rightarrow x \in (-\infty, -1] \cup [1-\sqrt{2}, 1+\sqrt{2}] \cup [3, \infty)$$

$$\therefore \alpha + \beta + \gamma + \delta = 4$$

14. Let $X = \{x \in \mathbb{N} : 1 \leq x \leq 19\}$ and for some $a, b \in \mathbb{R}$,

$Y = \{ax + b : x \in X\}$. If the mean and variance of the elements of Y are 30 and 750, respectively, then the sum of all possible values of b is

(1) 20 (2) 80

(3) 100 (4) 60

Ans. (4)

Sol. $\Sigma y_i = a \Sigma x_i + \Sigma b$

$$= a \times (1 + 2 + \dots + 19) + 19b$$

$$\frac{\Sigma y_i}{19} = \frac{a \times 19 \times 20}{2 \times 19} + b$$

$$30 = 10a + b \dots (1)$$

$$\text{Variance of } X = \frac{\sum x_i^2}{19} - \left(\frac{\sum x_i}{19} \right)^2$$

$$= \frac{19 \times 20 \times 39}{19 \times 6} - (10)^2 = 30$$

Variance of $Y = a^2$ (variance of X)

$$750 = a^2 \times 30$$

$$a^2 = 25 \Rightarrow a = \pm 5$$

$$\text{if } a = +5 \Rightarrow b = 30 - 50 = -20 \quad \dots \text{from (i)}$$

$$\text{if } a = -5 \Rightarrow b = 30 + 50 = 80 \quad \dots \text{from (i)}$$

$$\text{sum of values of } b = 80 - 20 = 60$$

options (4)

15. Consider the following three statements for the function $f: (0, \infty) \rightarrow \mathbb{R}$ defined by

$$f(x) = |\log_e x| - |x - 1| :$$

(I) f is differentiable at all $x > 0$.

(II) f is increasing in $(0, 1)$.

(III) f is decreasing in $(1, \infty)$.

Then.

(1) All (I), (II) and (III) are TRUE.

(2) Only (I) is TRUE.

(3) Only (II) and (III) are TRUE.

(4) Only (I) and (III) are TRUE.

Ans. (4)

Sol. $f(x) = |\ln x| - |x - 1|$

$$= \begin{cases} \ln x - (x - 1) & x \geq 1 \\ -\ln x + (x - 1) & 0 < x < 1 \end{cases}$$

$$= \begin{cases} \ln x - x + 1 & x \geq 1 \\ -\ln x + x - 1 & 0 < x < 1 \end{cases}$$

$$f'(x) = \begin{cases} \frac{1}{x} - 1 & x \geq 1 \\ -\frac{1}{x} + 1 & 0 < x < 1 \end{cases}$$

$$f'(1^+) = f'(1^-) = 0 \Rightarrow f(x) \text{ is differentiable } \forall x > 0$$

$$f'(x) < 0 \quad \forall x > 1$$

$$f'(x) < 0 \quad \forall 0 < x < 1$$

$$\Rightarrow f(x) \text{ is decreasing } \forall x \in (0, \infty)$$

Option (4)

16. Let $y = y(x)$ be a differentiable function in the interval $(0, \infty)$ such that $y(1) = 2$.

$$\text{and } \lim_{t \rightarrow x} \left(\frac{t^2 y(x) - x^2 y(t)}{x - t} \right) = 3 \quad \text{for each } x > 0.$$

Then $2y(2)$ is equal to

(1) 18

(2) 23

(3) 27

(4) 12

Ans. (2)

Sol. $\lim_{t \rightarrow x} \frac{2t f(x) - x^2 f'(t)}{-1} = 3$

$$x^2 f'(x) - 2x f(x) = 3$$

$$\frac{dy}{dx} - \frac{2y}{x} = \frac{3}{x^2}$$

$$\text{I.F.} = e^{-\int \frac{2}{x} dx} = e^{-2 \log_e x} = 1/x^2$$

$$y \cdot \frac{1}{x^2} = \int \frac{3}{x^4} dx$$

$$\frac{y}{x^2} = -\frac{1}{x^3} + c \Rightarrow y = cx^2 - \frac{1}{x} = f(x)$$

$$f(1) = 2 = c - 1 \Rightarrow c = 3$$

$$f(x) = 3x^2 - \frac{1}{x}$$

$$f(2) = 12 - \frac{1}{2} \Rightarrow 2f(2) = 23$$

17. Let f be a function such that $3f(x) + 2f\left(\frac{m}{19x}\right) = 5x$,

$$x \neq 0, \text{ where } m = \sum_{i=1}^9 (i)^2. \text{ Then } f(5) - f(2) \text{ is equal}$$

to

(1) -9

(2) 36

(3) 18

(4) 9

Ans. (3)

Sol. $m = \frac{9 \times 10 \times 19}{6} = 15 \times 19$

$$3f(x) + 2f\left(\frac{15}{x}\right) = 5x$$

Replace x by $\frac{15}{x}$

$$3f\left(\frac{15}{x}\right) + 2f(x) = \frac{75}{x}$$

$$9f(x) - 4f(x) = 15x - \frac{150}{x}$$

$$5f(x) = 15x - \frac{150}{x}$$

$$f(x) = 3x - \frac{30}{x}$$

$$f(5) = 15 - \frac{30}{5} = 9$$

$$f(2) = 6 - 15 = -9$$

$$f(5) - f(2) = 18$$

- 18.** The smallest positive integral value of a , for which all the roots of $x^4 - ax^2 + 9 = 0$ are real and distinct, is equal to

- (1) 9 (2) 3
(3) 4 (4) 7

Ans. (4)

Sol. $x^4 - ax^2 + 9 = 0 \quad \dots(1)$

let $x^2 = t$

$$t^2 - at + 9 = 0 \quad \dots(2)$$

for roots of equation (1) to be real & distinct roots of equation (2) must be positive & distinct.

(i) $D > 0 \Rightarrow a^2 - 36 > 0 \Rightarrow a \in (-\infty, -6) \cup (6, \infty)$

(ii) $\frac{-b}{2a} > 0 \Rightarrow \frac{a}{2} > 0 \Rightarrow a > 0$

(iii) $f(0) > 0 \Rightarrow 9 > 0 \Rightarrow a \in \mathbb{R}$

By (i) $\cap$ (ii) $\cap$ (iii)

$\therefore a \in (6, \infty)$

$\therefore$ least integral value of a is 7

- 19.** Let $\vec{a} = 2\hat{i} - 5\hat{j} + 5\hat{k}$ and $\vec{b} = \hat{i} - \hat{j} + 3\hat{k}$. If $\vec{c}$ is a vector such that

$$2(\vec{a} \times \vec{c}) + 3(\vec{b} \times \vec{c}) = \vec{0} \quad \text{and} \quad (\vec{a} - \vec{b}) \cdot \vec{c} = -97, \text{ then}$$

$$|\vec{c} \times \vec{k}|^2 \text{ is equal to}$$

- (1) 193 (2) 233
(3) 218 (4) 205

Ans. (3)

Sol. $2(\vec{a} \times \vec{c}) + 3(\vec{b} \times \vec{c}) = \vec{0}$

$$\Rightarrow (2\vec{a} + 3\vec{b}) \times \vec{c} = \vec{0} \Rightarrow \vec{c} = \lambda(2\vec{a} + 3\vec{b})$$

$$\Rightarrow \vec{c} = \lambda(7\hat{i} - 13\hat{j} + 19\hat{k})$$

$$\text{Now } (\vec{a} - \vec{b}) \cdot \vec{c} = \lambda(7 + 52 + 38) + 97\lambda = -97$$

$$\Rightarrow \lambda = -1$$

$$\text{Now } \vec{c} = -7\hat{i} + 13\hat{j} - 19\hat{k}$$

$$\Rightarrow \vec{c} \times \vec{k} = -7\hat{j} + 13\hat{i} \Rightarrow |\vec{c} \times \vec{k}|^2 = 7^2 + 13^2 = 218$$

- 20.** Let $[t]$ denote the greatest integer less than or equal to t . If the function

$$f(x) = \begin{cases} b^2 \sin\left(\frac{\pi}{2} \left[\frac{\pi}{2} (\cos x + \sin x) \cos x \right] \right) & , x < 0 \\ \frac{\sin x - \frac{1}{2} \sin 2x}{x^3} & , x > 0 \\ a & , x = 0 \end{cases}$$

is continuous at $x = 0$, then $a^2 + b^2$ is equal to

- (1) $\frac{5}{8}$ (2) $\frac{9}{16}$
(3) $\frac{3}{4}$ (4) $\frac{1}{2}$

Ans. (3)

Sol. $f(0) = a$

$$\text{RHL} = \lim_{x \rightarrow 0^+} \frac{\sin x (1 - \cos x)}{x^3} = \frac{1}{2}$$

$$\text{LHL} = \lim_{x \rightarrow 0^-} \left(b^2 \sin \left[\frac{\pi}{2} \left[\frac{\pi}{2} (\sin x + \cos x) \cos x \right] \right] \right) = b^2$$

$$\therefore a = \frac{1}{2} \text{ \& } b^2 = \frac{1}{2}$$

$$\text{so } (a^2 + b^2) = \frac{1}{4} + \frac{1}{2} = \frac{3}{4}$$

SECTION-B

21. If $f(x)$ satisfies the relation $f(x) = e^x + \int_0^1 (y + xe^x) f(y) dy$, then $e + f(0)$ is equal to _____.

Ans. (2)

Sol. $f(x) = e^x + \int_0^1 yf(y)dy + xe^x \int_0^1 f(y)dy$
 $f(x) = e^x + A + Bxe^x$
 $A = \int_0^1 yf(y) dy = \int_0^1 y(A + e^y + By e^y) dy$
 $A = \frac{A}{2} + (0 - (-1)) + B(e - 1)$
 $\frac{A}{2} + B(1 - e) = 1$
 $B = \int_0^1 f(y) dy$
 $B = \int_0^1 (e^y + A + By e^y) dy$
 $B = (e - 1) + A + B(0 - (-1))$
 $B = e - 1 + A + B \Rightarrow A = 1 - e$
 $f(x) = e^x + A + Bxe^x$
 $f(0) = 1 + A = 1 - e + 1 = 2 - e$
 $e + f(0) = 2$

22. Let (h, k) lie on the circle $C : x^2 + y^2 = 4$ and the point $(2h + 1, 3k + 2)$ lie on an ellipse with eccentricity e . Then the value of $\frac{5}{e^2}$ is equal to _____.

Ans. (9)

Sol. Let $P \equiv (2\cos\theta, 2\sin\theta)$
 $\therefore$ coordinates of $Q = (4\cos\theta + 1, 6\sin\theta + 3)$
 $\therefore$ locus of Q is $\left(\frac{x-1}{4}\right)^2 + \left(\frac{y-3}{6}\right)^2 = 1$
 $\therefore e^2 = 1 - \frac{16}{36} = \frac{5}{9}$
 $\therefore \boxed{\frac{5}{e^2} = 9}$

23. Let $z = (1 + i)(1 + 2i)(1 + 3i) \dots (1 + ni)$, where $i = \sqrt{-1}$. If $|z|^2 = 44200$, then n is equal to _____.

Ans. (5)

Sol. $|z|^2 = 2^3 \cdot 5^2 \cdot 13 \cdot 17$
 $\prod_{r=1}^n (1+r^2) = 2^3 \cdot 5^2 \cdot 13 \cdot 17 = (2) \cdot (5) \cdot (2 \cdot 5) \cdot (17) \cdot (2 \cdot 13) = 2 \cdot 5 \cdot 10 \cdot 17 \cdot 26$
 $\therefore n = 5$

24. Let S be a set of 5 elements and $P(S)$ denote the power set of S . Let E be an event of choosing an ordered pair (A, B) from the set $P(S) \times P(S)$ such that $A \cap B = \emptyset$. If the probability of the event E is $\frac{3^p}{2^q}$, where $p, q \in \mathbb{N}$, then $p + q$ is equal to _____.

Ans. (15)

Sol. $S = \{a, b, c, d, e\}$
 $P(S)$ contains 32 elements
 both set A and set B are subsets of $P(S)$
 Every element has 4 choices

A	B
✓	✓
✓	x
x	✓
x	x

Favourable cases = 3^5

Total cases = 4^5

$$P = \frac{3^5}{4^5} = \frac{3^5}{2^{10}}$$

$$m = 5, n = 10$$

$$m + n = 15$$

25. The number of elements in the set $\left\{ x \in [0, 180^\circ] : \tan(x + 100^\circ) = \tan(x + 50^\circ) \tan x \tan(x - 50^\circ) \right\}$ is _____.

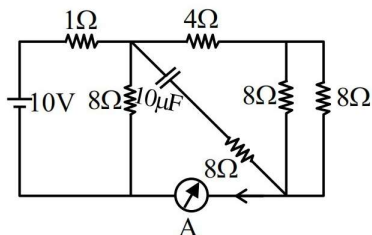
Ans. (4)

Sol. $\frac{\tan(x + 100^\circ)}{\tan x} = \tan(x + 50^\circ) \tan(x - 50^\circ)$
 $\frac{\sin(x + 100^\circ) \cos x}{\cos(x + 100^\circ) \sin x} = \frac{\sin(x + 50^\circ) \sin(x - 50^\circ)}{\cos(x + 50^\circ) \cos(x - 50^\circ)}$
 Apply C & D
 $\frac{\sin(2x + 100^\circ)}{\sin 100^\circ} = \frac{\cos 100^\circ}{-\cos 2x}$
 $2\sin(2x + 100^\circ) \cos 2x + \sin 200^\circ = 0$
 $\sin(4x + 100^\circ) + \sin 100^\circ + \sin 200^\circ = 0$
 $\sin(4x + 100^\circ) = -2\sin 150^\circ \cos 50^\circ$
 $\sin(4x + 100^\circ) = -\cos 50^\circ = \sin(-40^\circ)$
 $\therefore 4x + 100^\circ = n\pi + (-1)^n \cdot (-40^\circ)$
 $x = \frac{n\pi + (-1)^{n+1}(40^\circ) - 100^\circ}{4}$
 $\therefore x = 30^\circ, 55^\circ, 120^\circ, 145^\circ$ in $(0, \pi)$
 $\therefore$ no. of solutions = 4

PHYSICS

SECTION-A

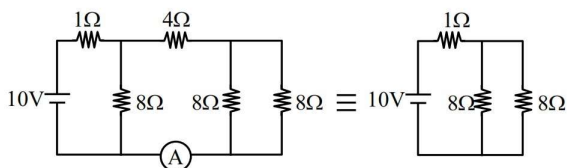
26. The reading of the ammeter (A) in steady state in the following circuit (assuming negligible internal resistance of the ammeter) is _____ A.



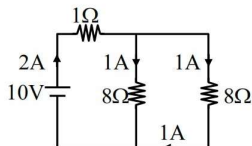
- (1) 2 (2) 1
(3) 1/2 (4) 0

Ans. (2)

Sol. In steady state



$$I = 2A$$



Ammeter reading is 1A.

27. In a vernier callipers, 50 vernier scale divisions are equal to 48 main scale divisions. If one main scale division = 0.05 mm, then the least count of the vernier callipers is _____ mm.
- (1) 0.002 (2) 0.05
(3) 0.02 (4) 0.005

Ans. (1)

Sol. $LC = 1 \text{ MSD} - 1 \text{ MSD} = 1 \text{ MSD} - \frac{48}{50} \text{ MSD}$

$$= \frac{2}{50} \text{ MSD} = \frac{2}{50} \times 0.05 \text{ mm} = 0.002 \text{ mm}$$

28. Five persons P_1, P_2, P_3, P_4 and P_5 recorded object distance (u) and image distance (v) using same convex lens having power +5D as (25,96), (30,62), (35,37), (45,35) and (50,32) respectively. Identify correct statement

- (1) Readings recorded by all persons are correct
(2) Reading recorded by P_3 persons are incorrect
(3) Reading recorded by P_3 and P_2 persons are incorrect
(4) Reading recorded by P_4 and P_5 persons are incorrect

Ans. (2)

Sol. $P = +5D$

$$\frac{1}{f} = 5 \Rightarrow f = 20 \text{ cm}$$

$\Rightarrow$ If object is between f & $2f$ image will be beyond $2f$ & magnified.

$\Rightarrow$ If object is beyond $2f$, image will be between f & $2f$ & diminished.

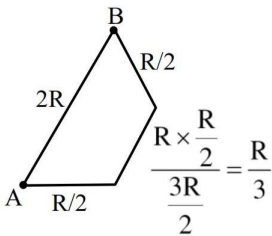
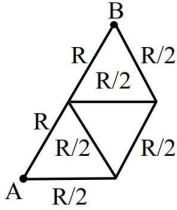
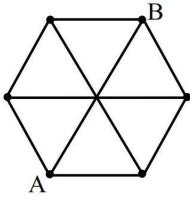
Hence reading of P_3 are incorrect.

29. A regular hexagon is formed by six wires each of resistance $r \Omega$ and the corners are joined to centre by wires of same resistance. If the current enters at one corner and leaves at the opposite corner, the equivalent resistance of the hexagon between the two opposite corners will be

- (1) $\frac{4}{5}r$ (2) $\frac{5}{8}r$
(3) $\frac{3}{4}r$ (4) $\frac{3}{5}r$

Ans. (1)

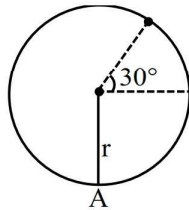
Sol.



$$R_{eq} = \frac{2R \times \frac{4R}{3}}{2R + \frac{4R}{3}} = \frac{8R^2}{10R} = \frac{4}{5}R$$

30. In case of vertical circular motion of a particle by a thread of length r if the tension in the thread is zero at an angle 30° shown in figure, the velocity at the bottom point (A) of the circular path is

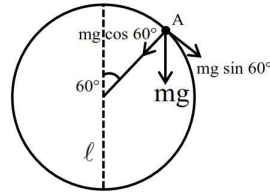
(g = gravitational acceleration)



- (1) $\sqrt{5gr}$ (2) $\sqrt{\frac{7}{2}gr}$
 (3) $\sqrt{4gr}$ (4) $\sqrt{\frac{5}{2}gr}$

Ans. (2)

Sol.



$$T + mg \cos 60^\circ = \frac{mV^2}{l}$$

$$T = 0$$

$$V^2 = \frac{g\ell}{2} \text{ here } V \text{ is the speed at point A}$$

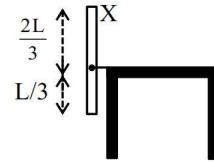
M.E.C.

$$\frac{1}{2}mu^2 = mg(\ell + \ell \cos 60^\circ) + \frac{1}{2}mV^2$$

$$u^2 = 3g\ell + \frac{g\ell}{2}$$

$$u = \sqrt{\frac{7g\ell}{2}}$$

31. A thin uniform rod (X) of mass M and length L is pivoted at a height $\left(\frac{L}{3}\right)$ as shown in the figure. The rod is allowed to fall from a vertical position and lie horizontally on the table. The angular velocity of this rod when it hits the table top, is _____. (g = gravitational acceleration)



- (1) $\sqrt{\frac{3}{2} \frac{g}{L}}$ (2) $\frac{3}{\sqrt{2}} \sqrt{\frac{g}{L}}$
 (3) $\frac{1}{\sqrt{2}} \sqrt{\frac{g}{L}}$ (4) $\sqrt{\frac{3g}{L}}$

Ans. (4)

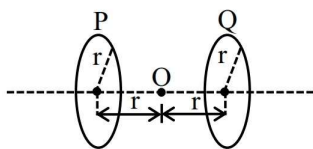
$$\text{Sol. } mg \frac{\ell}{6} = \frac{1}{2} I \omega^2$$

$$\text{Here } I = \frac{m\ell^2}{12} + \frac{m\ell^2}{36} = \frac{m\ell^2}{9}$$

$$mg \frac{\ell}{6} = \frac{m\ell^2}{18} \omega^2 \Rightarrow \omega^2 = \frac{3g}{\ell}$$

$$\omega = \sqrt{\frac{3g}{\ell}}$$

32.



Two identical circular loops P and Q each of radius r are lying in parallel planes such that they have common axis. The current through P and Q are I and $4I$ respectively in clockwise direction as seen from O. The net magnetic field at O is:

- (1) $\frac{3\mu_0 I}{4\sqrt{2}r}$ toward P
- (2) $\frac{\mu_0 I}{4\sqrt{2}r}$ toward P
- (3) $\frac{\mu_0 I}{4\sqrt{2}r}$ towards Q
- (4) $\frac{3\mu_0 I}{4\sqrt{2}r}$ towards Q

Ans. (4)

Sol. $B_{\text{net}} = B_1 - B_2$

$$\begin{aligned}
 & \text{Diagram: } Q \leftarrow B_2 \quad O \quad B_1 \rightarrow P \\
 & = \frac{4\mu_0 i R^2}{2(R^2 + R^2)^{3/2}} - \frac{\mu_0 i R^2}{2(R^2 + R^2)^{3/2}} \\
 & = \frac{3\mu_0 i}{4\sqrt{2}R}
 \end{aligned}$$

33. When a light of a given wavelength falls on a metallic surface the stopping potential for photoelectrons is 3.2 V. If a second light having wavelength twice of first light is used, the stopping potential drops to 0.7 V. The wavelength of first light is _____ m.

$$(h = 6.63 \times 10^{-34} \text{ J.s, } e = 1.6 \times 10^{-19} \text{ C, } c = 3 \times 10^8 \text{ m/s})$$

- (1) 2.9×10^{-8}
- (2) 2.2×10^{-8}
- (3) 3.1×10^{-7}
- (4) 2.5×10^{-7}

Ans. (4)

Sol. $q.(3.2) = \frac{hc}{\lambda} - \phi \quad \dots(1)$

$$q.(0.7) = \frac{hc}{2\lambda} - \phi \quad \dots(2)$$

$$\text{Eq. (1)} - \text{Eq. (2)}$$

$$q.(2.5) = \frac{hc}{2\lambda}$$

$$2.5 = \left(\frac{hc}{e} \right) \left(\frac{1}{2\lambda} \right)$$

$$2.5 = \frac{12400}{2(\lambda)}$$

$$\lambda = \frac{12400}{5} \text{ \AA}$$

$$\lambda = 2480 \text{ \AA}$$

$$\lambda = 2.48 \times 10^{-7} \text{ m}$$

34. The fifth harmonic of a closed organ pipe is found to be in unison with the first harmonic of an open pipe. The ratio of lengths of closed pipe to that of the open pipe is $5/x$. The value of x is _____.

- (1) 4
- (2) 2
- (3) 1
- (4) 3

Ans. (2)

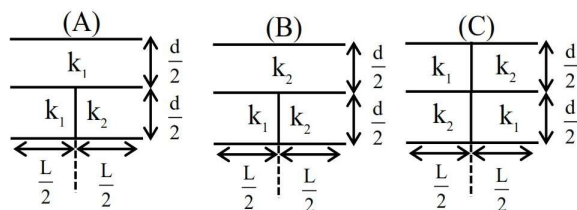
Sol. $f_{5 \text{ closed}} = f_{1 \text{ open}}$

$$\frac{5v}{4L_{\text{closed}}} = \frac{v}{2L_{\text{open}}}$$

$$\frac{L_{\text{closed}}}{L_{\text{open}}} = \frac{5}{2}$$

$$x = 2$$

35. Three parallel plate capacitors each with area A and separation d are filled with two dielectric (k_1 and k_2) in the following fashion. Which of the following is true? ($k_1 > k_2$)



$$(1) C_B > C_C > C_A$$

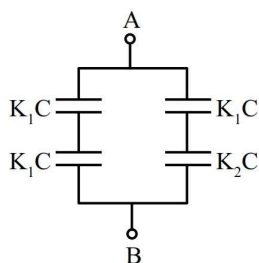
$$(2) C_C > C_B > C_A$$

$$(3) C_C > C_A > C_B$$

$$(4) C_A > C_C > C_B$$

Ans. (4)

Sol. For C_A :

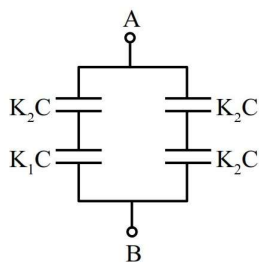


$$\text{Let } \frac{\epsilon_0 A}{d} = C$$

$$\therefore C_A = \frac{K_1 C}{2} + \frac{K_1 K_2 C}{K_1 + K_2}$$

$$= K_1 C \left[\frac{K_1 + 2K_2}{2(K_1 + K_2)} \right]$$

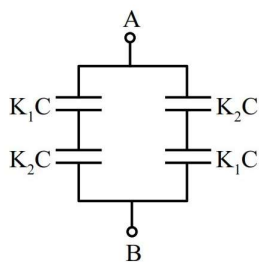
For C_B :



$$C_B = \frac{K_2 C}{2} + \frac{K_1 K_2 C}{K_1 + K_2}$$

$$= K_2 C \left[\frac{K_1 + 2K_2}{2(K_1 + K_2)} \right]$$

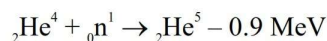
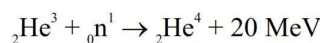
For C_C :



$$C_C = \frac{2K_1 K_2 C}{(K_1 + K_2)}$$

$$C_A > C_C > C_B$$

36. The binding energy for the following nuclear reactions are expressed in MeV.



If X_3 , X_4 , X_5 denote the stability of ${}_2\text{He}^3$, ${}_2\text{He}^4$ and ${}_2\text{He}^5$, respectively, then the correct order is :

(1) $X_4 > X_5 > X_3$

(2) $X_4 = X_5 = X_3$

(3) $X_4 > X_5 < X_3$

(4) $X_4 < X_5 < X_3$

Ans. (1)

Sol. $BE_{\text{He}^4} - BE_{\text{He}^3} = 20 \text{ MeV} \quad \dots(1)$

$$BE_{\text{He}^5} - BE_{\text{He}^4} = -0.9 \text{ MeV} \quad \dots(2)$$

From eq (1) & (2)

$$BE_{\text{He}^4} > BE_{\text{He}^5} > BE_{\text{He}^3}$$

$$X_4 > X_5 > X_3$$

37. A cubical block of density $\rho_b = 600 \text{ kg/m}^3$ floats in a liquid of density $\rho_c = 900 \text{ kg/m}^3$. If the height of block is $H = 8.0 \text{ cm}$ then height of the submerged part is _____ cm.

(1) 7.3

(2) 4.3

(3) 6.3

(4) 5.3

Ans. (4)

Sol. $Mg = F_b$

$$dAHg = \rho Ahg$$

$$600 \times 8 \text{ cm} = 900 \times h$$

$$h = \frac{16}{3} \text{ cm}$$

$$h = 5.3 \text{ cm}$$

38. A moving coil galvanometer of resistance 100Ω shows a full scale deflection for a current of 1 mA . The value of resistance required to convert this galvanometer into an ammeter, showing full scale deflection for a current of 5 mA , is _____ Ω

(1) 25

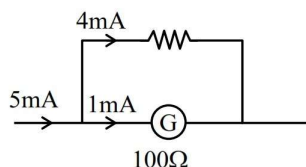
(2) 10

(3) 0.5

(4) 2.5

Ans. (1)

Sol.



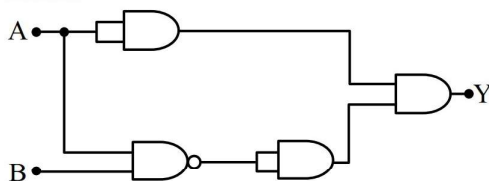
$$G = 100 \Omega$$

$$i_g = 1\text{mA}$$

$$i = 5\text{mA}$$

$$r_s = \frac{G}{\left(\frac{i}{i_g} - 1\right)} = \frac{100}{\left(\frac{5}{1} - 1\right)} = 25\Omega$$

39. Identify the correct truth table of the given logical circuit.



(1)

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

(2)

A	B	Y
0	0	1
0	1	0
1	0	1
1	1	0

(3)

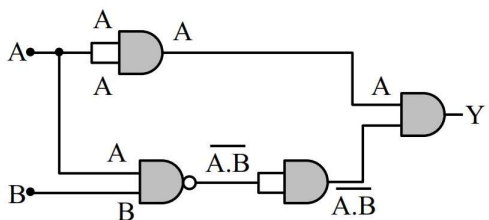
A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

(4)

A	B	Y
0	0	0
0	1	0
1	0	1
1	1	0

Ans. (4)

Sol.



$$y = A \cdot \overline{A \cdot B}$$

$$= A \cdot (\overline{A} + \overline{B}) = 0 + A\overline{B}$$

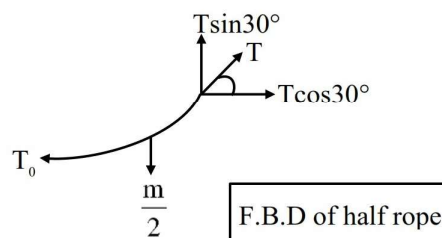
A	B	Y
0	0	0
0	1	0
1	0	1
1	1	0

40. A flexible chain of mass m hangs between two fixed points at the same level. The inclination of the chain with the horizontal at the two points of support is 30° . Considering the equilibrium of each half of the chain, the tension of the chain at the lowest point is ____.

(1) $\frac{\sqrt{3}}{2}mg$ (2) $\frac{1}{2}mg$ (3) mg (4) $\sqrt{3}mg$

Ans. (1)

Sol.



$$T \sin 30^\circ = \frac{m}{2}g$$

$$T \cos 30^\circ = T_0$$

$$\tan 30^\circ = \frac{mg}{2T_0}$$

$$T_0 = \frac{\sqrt{3}}{2}mg$$

41. A point source is kept at the center of a spherically enclosed detector. If the volume of the detector increased by 8 times, the intensity will

(1) increase by 8 times (2) increase by 64 times
(3) decrease by 8 times (4) decrease by 4 times

Ans. (4)

Sol. $V \rightarrow 8V \Rightarrow R \rightarrow 2R$

$$\Rightarrow A \rightarrow 4A$$

$$\Rightarrow I \rightarrow \frac{I_0}{4}$$

42. In the Young's double slit experiment the intensity produced by each one of the individual slits is I_0 . The distance between two slits is 2 mm. The distance of screen from slits is 10 m. The wavelength of light is $6000 \text{ \AA}$. The intensity of light on the screen in front of one of the slits is ____.

- (1) $2I_0$ (2) I_0 (3) $\frac{I_0}{2}$ (4) $4I_0$

Ans. (2)

Sol. $d = 2\text{mm}$

$D = 10 \text{ m}$

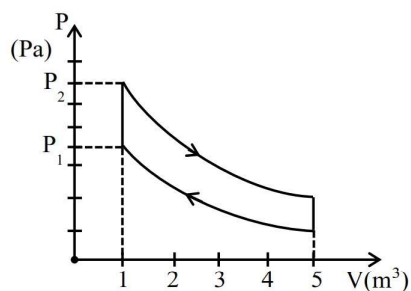
$\lambda = 6000 \text{ \AA}$

$$y = \frac{d}{2} \text{ (in front of one slit)}$$

$$I = 4I_0 \cos^2 \left(\frac{2\pi}{\lambda} \cdot \frac{y}{D} d \right)$$

$$\Rightarrow I = I_0$$

43. 10 mole of an ideal gas is undergoing the process shown in the figure. The heat involved in the process from P_1 to P_2 is α Joule ($P_1 = 21.7 \text{ Pa}$ and $P_2 = 30 \text{ Pa}$, $C_v = 21 \text{ J/K.mol}$, $R = 8.3 \text{ J/mol.K}$). The value of α is ____.



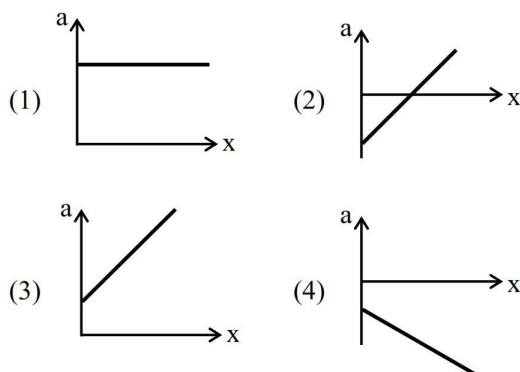
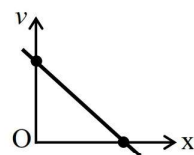
- (1) 24 (2) 15 (3) 21 (4) 28

Ans. (3)

Sol. $\Delta Q = nC_v \Delta T$ (isochoric)

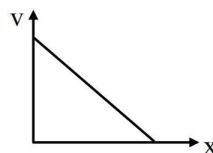
$$\begin{aligned} &= \frac{C_v}{R} \cdot nR\Delta T = \frac{C_v}{R} (P_2 - P_1) V \\ &= \frac{21}{8.3} \times (30 - 21.7) \times 1 = 21 \text{ J} \end{aligned}$$

44. The velocity (v) – Distance (x) graph is shown in figure. Which graph represents acceleration (a) versus distance (x) variation of this system?



Ans. (2)

Sol.



Eq. of V vs x from graph

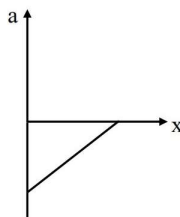
$$V = C_1 - C_2 x$$

$$a = V \frac{dV}{dx}$$

$$= (C_1 - C_2 x) \times -C_2$$

$$a = C_2^2 x - C_1 C_2$$

$\therefore$ graph is straight line +ve slope -ve intercept



45. Distance between an object and three times magnified real image is 40 cm. The focal length of the mirror used is _____ cm.

- (1) $-15/2$ (2) -10
(3) -20 (4) -15

Ans. (4)

Sol. $m = -3 = \frac{v}{u}$

$$v = -3u$$

$$|v| - |u| = 40$$

$$u = 20 \text{ cm}$$

$$v = 60 \text{ cm}$$

$$\therefore \frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{-60} + \frac{1}{-20} = \frac{1}{f}$$

$$f = -15 \text{ cm}$$

SECTION-B

46. When 300 J of heat given to an ideal gas with $C_p = \frac{7}{2}R$ its temperature raises from 20°C to 50°C keeping its volume constant. The mass of the gas is (approximately) _____ g. ($R = 8.314 \text{ J/mol.K}$).

Ans. (481)

Sol. $C_v = C_p - R = \frac{5}{2}R$

$$\Delta Q = nC_v \Delta T$$

$$300 = n \times \frac{5}{2} \times 8.314 \times 30$$

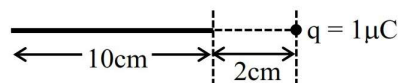
$$n = 0.48$$

$$\frac{m}{M} = 0.48$$

We cannot find mass (m) because molar mass (M) not given.

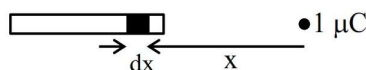
47. A point charge $q = 1 \mu\text{C}$ is located at a distance 2 cm from one end of a thin insulating wire of length 10 cm having a charge $Q = 24 \mu\text{C}$, distributed uniformly along its length, as shown in figure. Force between q and wire is _____ N.

(Use $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ N.m}^2 / \text{C}^2$)



Ans. (90)

Sol.



$$F = \int dF = \int_{2\text{cm}}^{12\text{cm}} \frac{kq\lambda dx}{x^2} = kq\lambda \left(\frac{1}{2 \times 10^{-2}} - \frac{1}{12 \times 10^{-2}} \right)$$

$$F = (9 \times 10^9)(10^{-6}) \left(\frac{24 \times 10^{-6}}{10^{-1}} \right) \left(\frac{5}{12} \right) \times 10^2$$

$$= 9 \times 24 \times \frac{5}{12} = 90 \text{ N}$$

48. In a meter bridge experiment to determine the value of unknown resistance, first the resistances 2Ω and 3Ω are connected in the left and right gaps of the bridge and the null point is obtained at a distance l cm from the left. Now when an unknown resistance $x \Omega$ is connected in parallel to 3Ω resistance, the null point is shifted by 10 cm to the right of wire. The value of unknown resistance x is _____ Ω .

Ans. (6)

Sol. In case I

$$\frac{2}{3} = \frac{\ell}{(100 - \ell)} \dots\dots(1)$$

$$\ell = 40 \text{ cm}$$

In case II

$$\frac{2}{R} = \frac{\ell + 10}{100 - (\ell + 10)}$$

Put $\ell = 40 \text{ cm}$ & solve

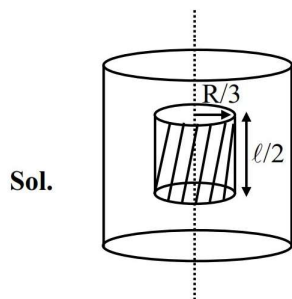
$$R = 2\Omega$$

$$\therefore \frac{3x}{3 + x} = 2$$

$$x = 6\Omega$$

49. A uniform solid cylinder of length L and radius R has moment of inertia about its axis equal to I_1 . A small co-centric cylinder of length $L/2$ and radius $R/3$ carved from this cylinder has moment of inertia about its axis equals to I_2 . The ratio I_1 / I_2 is _____.

Ans. (162)



Original mass (M)

The removed mass (m)

$$m = \rho \times \pi \left(\frac{R}{3} \right)^2 \times \frac{L}{2}$$

$$= \frac{\rho \pi R^2 L}{18} = \frac{M}{18}$$

$$I' = \frac{1}{2} \cdot \frac{M}{18} \cdot \frac{R^2}{9} = \frac{1}{324} MR^2$$

$$\frac{I}{I'} = \frac{\frac{1}{2} MR^2}{\frac{1}{324} MR^2} = 162$$

50. A soap bubble of surface tension 0.04 N/m is blown to a diameter of 7 cm . If $(15000 - x) \mu\text{J}$ of work is done in blowing it further to make its diameter 14 cm , then the value of x is _____.

$$(\pi = 22/7)$$

Ans. (11304)

Sol. $W = \Delta u$

$$= S \times (8\pi r_2^2 - 8\pi r_1^2)$$

$$= 0.04 \times 2 \times \frac{22}{7} (147) \times 10^{-4}$$

$$W = 3696 \times 10^{-6} \text{ J}$$

$$3696 = 15000 - x$$

$$x = 11304 \mu\text{J}$$

CHEMISTRY

SECTION-A

51. Choose the **INCORRECT** statement

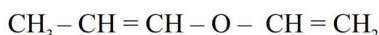
- (1) Among the isotopes of carbon, ^{13}C is a radioactive isotope.
- (2) Carbon exhibits negative oxidation states along with +4 and +2.
- (3) Carbon cannot exceed its covalency more than four.
- (4) CO_2 is the most acidic oxide among the dioxides of group of 14 elements.

Ans. (1)

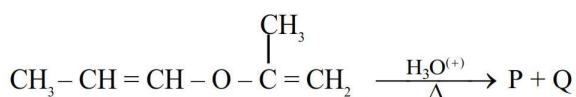
Sol. C^{13} is not radioactive

C^{14} is radioactive

52. The unsaturated ether on acidic hydrolysis produces carbonyl compounds as shown below:-

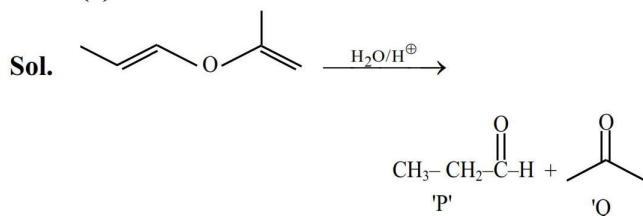


Based on this, predict the solution / reagent that will help to distinguish "P" and "Q" obtained in the following reaction.



- (1) Lucas reagent
- (2) 2,4-DNP reagent
- (3) Saturated NaHSO_3 solution
- (4) Fehling solution

Ans. (4)



'P' and 'Q' can be differentiated by Fehling's test.

P gives positive Fehling test

Q gives negative Fehling test

53. The number of possible tripeptides formed involving alanine (ala), glycine (gly) and valine (val), where no amino acid has been used more than once is :

- (1) 6
- (2) 3
- (3) 4
- (4) 8

Ans. (1)

Sol. Gly ala val

Gly val ala

Val gly ala

Val ala gly

Ala val gly

Ala gly val

Total tri peptides = 6

54. Two liquids A and B form an ideal solution at temperature T K. At T K, the vapour pressures of pure A and B are 55 and 15 kNm^{-2} respectively. What is the mole fraction of A in solution of A and B in equilibrium with a vapour in which the mole fraction of A is 0.8 ?

- (1) 0.5217
- (2) 0.480
- (3) 0.663
- (4) 0.340

Ans. (1)

Sol.
$$\frac{Y_A}{Y_B} = \frac{P_A^0}{P_B^0} \cdot \frac{X_A}{X_B}$$

$$\frac{0.8}{0.2} = \frac{55}{15} \times \frac{X_A}{X_B}$$

$$\frac{X_A}{X_B} = \frac{60}{55} = \frac{12}{11}$$

$$X_A = \frac{12}{23} = 0.5217$$

Ans. (3)



$t = 0$ 1 mole 4 mole

t_f 0 2 mole 1 mole 1 mole

$[\text{OH}] = 1 \text{ M}$

$$[\text{Cl}] = \frac{1}{2} \text{ M}$$

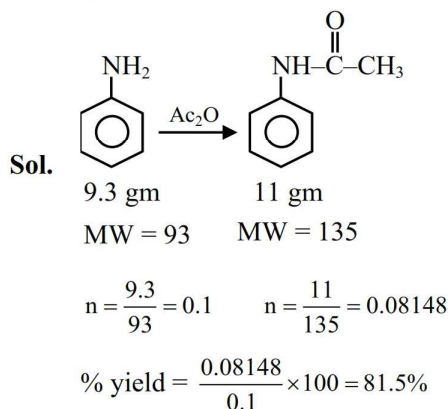
$$[\text{ClO}] = \frac{1}{2} \text{ M}$$

- 59.** A student has planned to prepare acetanilide from aniline using acetic anhydride. The student has started from 9.3g of aniline. However, the student has managed to obtain 11 g of dry acetanilide.

The % yield of this reaction is :-

- (1) 81.5% (2) 97.5%
(3) 59.5% (4) 72.5%

Ans. (1)



- 60.** Find out the statements which are **not** true.

- A. Resonating structure with more number of covalent bonds and lesser charge separation are more stable.
- B. In electromeric effect, an unsaturated system shows + E effect with nucleophile and -E effect with electrophile.
- C. Inductive effect is responsible for high melting point, boiling point and dipole moment of polar compounds.
- D. The greater the number of alkyl groups attached to the doubly bonded carbon atoms, higher is the heat of hydrogenation.
- E. Stability of carbanion increases with the increase in s-character of the carbon carrying the negative charge.

Choose the **correct** answer from the options given below.

- (1) A, D & E only (2) B, D & E only
(3) A, C & D only (4) B & D only

Ans. (4)

Sol. Statement B & D are not true

- 61.** The correct order of C, N, O and F in terms of second ionisation potential is

- (1) $F < N < C < O$ (2) $C < O < N < F$
(3) $C < N < F < O$ (4) $C < F < N < O$

Ans. (2)

Sol. To compare second ionization potential configuration of mono-cation is observed

C^+	N^+	O^+	F^+
$[\text{He}] 2s^2 2p^1$	$[\text{He}] 2s^2 2p^2$	$[\text{He}] 2s^2 2p^3$ Half-filled stable.	$[\text{He}] 2s^2 2p^4$

2nd IE order

$O > F > N > C$

- 62.** In the Group analysis of cations, Ba^{2+} & Ca^{2+} are precipitated respectively as

- (1) sulphide & sulphide
(2) hydroxide & carbonate
(3) carbonate & carbonate
(4) chromate & sulphide

Ans. (3)

Sol. To identify Ba^{2+} & Ca^{2+}

Reagent $(\text{NH}_4)_2\text{CO}_3 + \text{NH}_4\text{Cl}$ is used BaCO_3 & CaCO_3 are obtained as precipitates

- 63.** The wavelength of spectral line obtained in the spectrum of Li^{2+} ion, when the transition takes place between two levels whose sum is 4 and difference is 2, is

- (1) $2.28 \times 10^{-7} \text{ cm}$
(2) $2.28 \times 10^{-6} \text{ cm}$
(3) $1.14 \times 10^{-7} \text{ cm}$
(4) $1.14 \times 10^{-6} \text{ cm}$

Ans. (4)

Sol. $n_1 \rightarrow$ lower energy level

$n_2 \rightarrow$ higher energy level

$$n_1 + n_2 = 4, \quad n_2 = 3$$

$$n_2 - n_1 = 2, \quad n_1 = 1$$

Rydberg's formula :

$$\frac{1}{\lambda} = R_H Z^2 \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right]$$

$$\frac{1}{\lambda} = R_H (3)^2 \left[\frac{1}{1^2} - \frac{1}{3^2} \right]$$

$$\frac{1}{\lambda} = 8R_H$$

$$\lambda = \frac{1}{8R_H}$$

$$\lambda = \frac{1}{8 \times 1.1 \times 10^5}$$

$$\lambda = \frac{1000}{8.8} \times 10^{-8} \text{ cm}$$

$$\lambda = 113.63 \times 10^{-8} \text{ cm}$$

$$\lambda \approx 1.14 \times 10^{-6} \text{ cm}$$

64. Given below are two statements :

Statement I : Cross aldol condensation between two different aldehydes will always produce four different products.

Statement II : When semicarbazide reacts with a mixture of benzaldehyde and acetophenone under optimum pH, it forms a condensation product with acetophenone only.

In the light of the above statements, choose the **correct** answer from the options given below :

- (1) Both Statement I and Statement II are false
- (2) Statement I is false but Statement II is true
- (3) Both Statement I and Statement II are true
- (4) Statement I is true but Statement II is false

Ans. (1)

Sol. Statement I : False

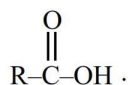
Cross aldol can give 2 or 4 products

Statement II : False

Benzaldehyde & Acetone both react with semi carbazide.

65. Given below are two statements :

Statement I : The dipole moment of R-CN is greater than R-NC and R-NC can undergo hydrolysis under acidic medium to produce



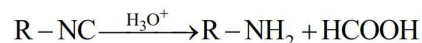
Statement II : R-CN hydrolyses under acidic medium to produce a compound which on treatment with SOCl_2 , followed by the addition of NH_3 gives another compound(x). This compound (x) on treatment with NaOCl/NaOH gives a product, that on treatment with $\text{CHCl}_3/\text{KOH}/\Delta$ produces R-NC

In the light of the above statements, choose the **correct** answer from the options given below :

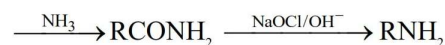
- (1) Both Statement I and Statement II are false
- (2) Both Statement I and Statement II are true
- (3) Statement I is true but Statement II is false
- (4) Statement I is false but Statement II is true

Ans. (4)

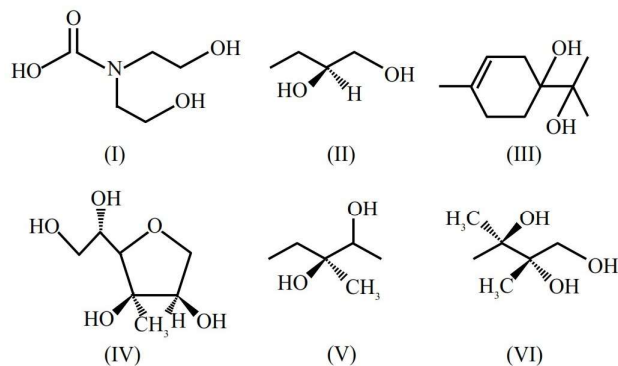
Sol. Statement I : False



Statement II : True



66. From the following, how many compounds contain at least one secondary alcohol ?



Choose the **correct** answer from the options given below :

- (1) Five (2) Three
(3) Four (4) two

Ans. (2)

Sol. II, IV & V are secondary alcohol.

67. The heat of atomisation of methane and ethane are 'x' kJ mol⁻¹ and 'y' kJ mol⁻¹ respectively. The longest wavelength (λ) of light capable of breaking the C-C bond can be expressed in SI unit as :

- (1) $\frac{hc}{1000} \left(\frac{y-6x}{4} \right)^{-1}$ (2) $\frac{N_A hc}{250(4y-6x)}$
(3) $\frac{N_A hc}{250(y-6x)}$ (4) $N_A hc \left(y - \frac{6x}{4} \right)^{-1}$

Ans. (2)

Sol. $\text{CH}_4(\text{g}) \rightarrow \text{C}(\text{g}) + 4\text{H}(\text{g}); \Delta_f H = x \text{ kJ/mole}$

$\text{C}_2\text{H}_6(\text{g}) \rightarrow 2\text{C}(\text{g}) + 6\text{H}(\text{g}); \Delta_f H = y \text{ kJ/mole}$

$$1000x = 4 \times \epsilon_{\text{C-H}}$$

$$1000y = 1 \times \epsilon_{\text{C-C}} + 6 \times \epsilon_{\text{C-H}}$$

$$\epsilon_{\text{C-C}} = \left[y - \frac{3x}{2} \right] \times 1000 = \frac{hc}{\lambda} \cdot N_A$$

$$(\lambda') \text{ wavelength of photon} = \frac{hc N_A}{[4y - 6x] \times 250}$$

68. At 298 K, the mole percentage of $\text{N}_2(\text{g})$ in air is 80%. Water is in equilibrium with air at a pressure of 10 atm. What is the mole fraction of $\text{N}_2(\text{g})$ in water at 298 K ? (K_H for N_2 is $6.5 \times 10^7 \text{ mm Hg}$)

- (1) 1.23×10^{-7} (2) 1.17×10^{-4}
(3) 9.35×10^5 (4) 9.35×10^{-5}

Ans. (4)

Sol. $P_{\text{N}_2} = K_H \cdot X_{\text{N}_2}$

$$P_{\text{N}_2} = 0.8 \times 10 = 8 \text{ atm}$$

$$8 \times 760 = 6.5 \times 10^7 \times X_{\text{N}_2}$$

$$X_{\text{N}_2} = \frac{8 \times 760}{6.5 \times 10^7}$$

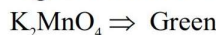
$$X_{\text{N}_2} = 9.35 \times 10^{-5}$$

69. "X" is an oxoanion of the lightest element of group 7 (in the periodic table). The metal is in +6 oxidation state in "X". The color of the potassium salt of X is

- (1) green (2) purple (3) yellow (4) orange

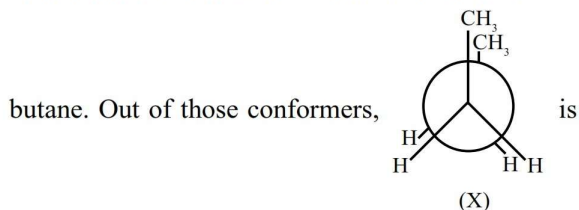
Ans. (1)

Sol. Lightest element of Group 7 $\Rightarrow \text{Mn}$

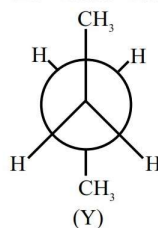


70. Given below are two statements :

Statement I : There are several conformers for n-



the least stable and most stable conformer is



Statement II : As the dihedral angle increases, torsional strain decreases from (X) to (Y).

In the light of the above statements, choose the **correct** answer from the options given below :

- (1) Both Statement I and Statement II are false
(2) Statement I is false but Statement II is true
(3) Statement I is true but Statement II is false
(4) Both Statement I and Statement II are true

Ans. (4)

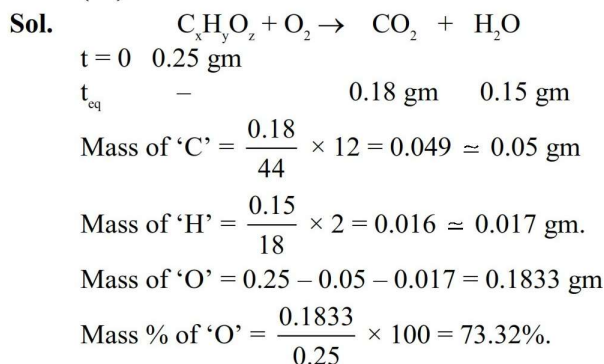
Sol. Both Statements are correct.

SECTION-B

71. 0.25 g of an organic compound "A" containing carbon, hydrogen and oxygen was analysed using the combustion method. There was an increase in mass of CaCl_2 tube and potash tube at the end of the experiment. The amount was found to be 0.15 g and 0.1837 g, respectively. The percentage of oxygen in compound A is _____. (Nearest integer)

(Given : molar mass in g mol^{-1} H : 1, C : 12, O : 16)

Ans. (73)



72. The half-life of ^{65}Zn is 245 days. After x days, 75% of original activity remained. The value of x in days is _____. (Nearest integer)

(Given : $\log 3 = 0.4771$ and $\log 2 = 0.3010$)

Ans. (102)

Sol. $t_{1/2} = \frac{\ell n 2}{K}$

$K = \frac{\ell n 2}{245}$

$t = \frac{1}{K} \ell n \frac{a_0}{a_t}$

$t_{25\%} = \frac{1}{K} \ell n \frac{4}{3}$

$t_{25\%} = \frac{1}{\ell n 2} \ell n \frac{4}{3}$

$t_{25\%} = 245 \frac{\ell n \frac{4}{3}}{\ell n 2} = 245 \left[\frac{2 \log 2 - \log 3}{\log 2} \right]$

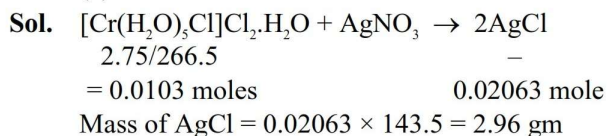
$= 245 \left[\frac{2 \times 0.3010 - 0.4771}{0.3010} \right] = 101.66$ day.

73. A chromium complex with a formula $\text{CrCl}_3 \cdot 6\text{H}_2\text{O}$ has a spin only magnetic moment value of 3.87 BM and its solution conductivity corresponds to 1 : 2 electrolyte. 2.75 g of the complex solution

was initially passed through a cation exchanger. The solution obtained after the process was reacted with excess of AgNO_3 . The amount of AgCl formed in the above process is _____ g. (Nearest integer)

[Given : Molar mass in g mol^{-1} Cr : 52; Cl : 35.5, Ag : 108, O : 16, H : 1]

Ans. (3)



74. Molar conductivity of a weak acid HQ of concentration 0.18 M was found to be 1/30 of the molar conductivity of another weak acid HZ with concentration of 0.02 of M. If λ_{Q}^0 happened to be

equal with λ_{Z}^0 , then the difference of the pK_a values of the two weak acids ($\text{pK}_a(\text{HQ}) - \text{pK}_a(\text{HZ})$) is _____. (Nearest integer).

[Given : degree of dissociation (α) $\ll 1$ for both weak acids, λ° : limiting molar conductivity of ions]

Ans. (2)

Sol. $K_a(\text{HQ}) = C_1 \alpha_1^2$ $\alpha_1 = \frac{\lambda_m(\text{HQ})}{\lambda_m^\infty(\text{HQ})}$

$K_a(\text{HZ}) = C_2 \alpha_2^2$ $\alpha_2 = \frac{\lambda_m(\text{HZ})}{\lambda_m^\infty(\text{HZ})}$

$\frac{K_a(\text{HQ})}{K_a(\text{HZ})} = \frac{C_1}{C_2} \cdot \left(\frac{\alpha_1}{\alpha_2} \right)^2 = \frac{0.18}{0.02} \cdot \left[\frac{\lambda_m(\text{HQ})}{\lambda_m(\text{HZ})} \right]^2$

$\frac{K_a(\text{HQ})}{K_a(\text{HQ})} = 9 \times \left(\frac{1}{30} \right)^2 = \frac{1}{100}$

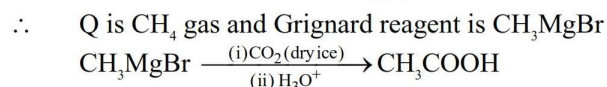
$\text{pK}_a(\text{HQ}) - \text{pK}_a(\text{HZ}) = 2$

75. Grignard reagent $\text{RMgBr}(\text{P})$ reacts with water and forms a gas (Q). One gram of Q occupies 1.4 dm^3 at STP. (P) on reaction with dry ice in dry ether followed by H_3O^+ forms a compound (Z). 0.1 mole of (Z) will weigh _____. (Nearest integer)

Ans. (6)

Sol. 1.4 dm^3 (or 1.4 mL) occupied by 1 gm

$\therefore$ Molecular weight of Q = $\frac{22.4}{1.4} = 16$



(Molecular weight 60)

$\therefore$ Weight of 0.1 mole of $\text{CH}_3\text{COOH} = 6$